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Speech production in stuttering: Impact of syllable frequency and word length on accuracy and fluency

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ABSTRACT

Developmental stuttering is a neurodevelopmental speech fluency disorder but its causes and underlying mechanisms have not yet been fully understood. Psycholinguistic studies have found lower non-word repetition accuracy in individuals who stutter, which has been proposed to reflect deficits in translating phonological plans into motor programs. At the interface between phonological and phonetic encoding, fluent speakers are thought to benefit from the retrieval of stored preassembled motor programs as evidenced by syllable frequency effects. However, it is unclear whether AWS also benefit from high-frequency syllables during speech production.

To systematically test phonetic encoding procedures in AWS, we designed a novel pseudoword repetition task with pseudowords of different lengths (2, 3 or 4 syllables) containing either high- or low-frequency first syllables to test for response accuracy and speech fluency in 19 AWS and 19 fluently speaking controls (FC). Results showed an increase in errors in both speaker groups and an increase in stuttering frequency in AWS with longer pseudowords. There was no significant effect of syllable frequency on speech fluency in AWS. However, for response accuracy there was a significant interaction between group and syllable frequency: Whereas FC showed the expected syllable frequency effect on response accuracy with more errors on pseudowords with LF first syllables, AWS produced *more* errors on pseudowords with high frequency first syllables. Based on Levelt and colleagues' psycholinguistic model, we interpreted the inverse syllable frequency effect as an indication of a disrupted interplay between the phonetic encoding routines of retrieving and assembling syllable motor programs in AWS whose stored motor programs might be more fragile.

1. Introduction

Developmental stuttering is a neurodevelopmental speech fluency disorder characterized by involuntary speech blocks, repetitions of sound and syllables, and prolongations of speech sounds. While its neurobiological mechanisms are not yet fully understood, multifactorial frameworks suggests that genetic, environmental, neurodevelopmental, and experiential factors may play a role (e.g. Smith and Weber, 2017). Theories associate the emergence of stuttering with the phonological/-phonetic encoding levels, where linguistic planning interfaces with motor programming, sensorimotor integration and execution (Civier et al., 2010, 2013; Howell, 2004; Max et al., 2004; Packman et al., 2007). According to Levelt's model of speech production, phonological and phonetic encoding are distinct stages in the process of converting thoughts into spoken words. Phonological encoding is about selecting and organizing the correct abstract speech units (phonemes) for speech, while phonetic encoding is about generating the motor commands needed to produce those speech units (phonemes). Thus, these speech processes coordinate the transitions between linguistic and motor planning (Levelt et al., 1999).

These processes are suggested to be particularly vulnerable in stuttering, as symptoms more likely occur when speaking long words, measured either by the number of phonemes or syllables (Brown, 1938a; Logan and Conture, 1995; Max et al., 2004), and when speaking low-frequency words compared to high-frequency words (e.g. Anderson, 2007; Hubbard and Prins, 1994; Newman and Bernstein Ratner, 2007).

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However, it should be noted that these length and frequency effects are not independent of one another: while longer and more complex words and structures tend to be less frequent, shorter and less complex structures are more frequent (Sigurd et al., 2004; Strauss et al., 2006). The current study investigates whether stuttering and speech errors reflect impaired coordination between linguistic planning and motor programming, drawing on Levelt's psycholinguistic model (1999) and Guenther's neurocomputational framework (Guenther, 2016).

1.1. Phonological encoding

According to Levelt et al.'s (1999) model of word production, phonological encoding begins with the retrieval of a given word's individual phonemes from long-term memory. While these phonemes are retrieved in parallel, their linear order is specified. During syllabification, these phonemes are assigned one by one to syllable-internal positions, forming abstract phonological syllables that are specified for suprasegmental properties such as stress assignment, allophonic variation, and assimilation. It is suggested that neural processing during phonological encoding involves posterior superior and middle temporal gyri during phonological code retrieval, and the left posterior inferior frontal gyrus (IFG) during syllabification (Indefrey, 2011; Indefrey and Levelt, 2004). Similarly, the neurocomputational model of speech production, GODIVA (Gradient Order Directions in Velocities of Articulators) suggests that phonological syllables exist within the pre-supplementary motor area and the inferior frontal sulcus, while their selection is mediated by the basal ganglia (Bohland et al., 2010). Furthermore, the planning of syllable sequences has been associated with the dorsal part of the IFG, the left inferior parietal lobe and the left supplementary motor area (SMA) to the planning of syllable sequences (Rong et al., 2018). The left IFG shows reduced grey matter volume in both children and adults who stutter (Beal et al., 2015; Chang et al., 2008; Kell et al., 2009), and lower activity in adults who stutter (AWS) compared to fluently speaking controls (Neef et al., 2016). AWS also have a reduced white matter connectivity in the left frontal aslant tract (Kronfeld-Duenias et al., 2016), which connects the IFG with the SMA. Stimulation of this tract during neurosurgery can induce stuttering (Kemerdere et al., 2016). In addition, AWS have been reported to exhibit reduced neural activity during speech planning compared to fluent controls (Chang et al., 2009). Some accounts suggest that planning difficulties at the phonological level could lead to stuttering. For example, this could occur when speakers attempt to execute an incomplete plan due to prolonged planning time (Howell, 2004). Within the GODIVA model, the possibility of stuttering due an impairment of the left cortical regions relevant for phonological encoding is discussed (Guenther, 2016).

1.2. Phonetic encoding

During phonetic encoding, abstract phonological syllables are converted into context-dependent phonetic syllables or motor programs that specify a set of motor routines to be executed by the articulators in a given phonological-phonetic environment (Levelt et al., 1999). Phonetic encoding encompasses several brain regions. The neural correlates of phonetic encoding described by the DIVA model are connected through a cortical-basal ganglia loop and a cortico-cortical loop, which involves the cerebellum and the thalamus. The cortical-basal ganglia loop encompasses the left ventral premotor cortex, the supplementary motor area, the putamen, the caudate nucleus, the globus pallidus and substantia nigra as well as nuclei of the thalamus (Guenther, 2016). Lesions to structures of this network, the SMA or the putamen, have been reported to lead to stuttering (Ackermann et al., 1996; Tani and Sakai, 2011; Theys et al., 2013). In addition, children and adults who stutter show reduced white matter integrity and aberrant functional activity and connectivity within the cortico-basal ganglia circuit (Chang and Zhu, 2013; Chang et al., 2015; Lu et al., 2010a; Metzger et al., 2018; Qiao et al., 2017; Watkins et al., 2008). Stuttering symptoms decrease to a certain level when pharmaceutical drugs that block the dopamine receptors within the basal ganglia are taken (Bothe et al., 2006; Maguire et al., 2020). Other accounts suggest that stuttering emerges due to timing difficulties to initiate the next or inhibit the previous phonetic syllable/speech motor program (Alm, 2004; Civier et al., 2010, 2013).

1.3. Syllable motor programs and the dual-route account of phonetic encoding

Both the DIVA model (Guenther, 2016) and the 'mental syllabary' theory (e.g., Levelt and Wheeldon, 1994) – an integral part of the Levelt et al. model of word production (Levelt et al., 1999) – converge on the assumption that (1) speakers utilize chunks of motor programs during the course of speech planning and (2), while motor programs can be as small as a single phoneme and as large as a word or even multisyllabic utterance, the most frequently or "typical sound chunk" (Meier and Guenther, 2023, p. 1319) is the syllable. According to the dual-route account of phonetic encoding (Cholin et al., 2011; Levelt et al., 1999; Levelt and Wheeldon, 1994), there are two pathways for generating phonetic syllables: a retrieval route and an assembly route. The retrieval route provides access to precompiled syllable motor programs that are stored in the hypothesized mental syllabary. The motor programs within the syllabary are organized in a spreading-activation network in which neighbouring units, – i.e., those that share segments in a specific position – are interconnected (Roelofs, 1997a, 1997b, 2002) and compete for selection: As soon as a phoneme is assigned to the onset position in a to-be-built syllable during syllabification, all motor programs in the syllabary that share segments in this position receive activation and spread activation to their neighbours. Eventually, the motor program that exceeds activation threshold and fully matches the addressing phonological segment string is verified via verification links (Levelt et al., 1999) and selected as target program. The alternative assembly route requires speakers to construct syllable programs by assembling phonetic segments or subsyllabic units such as onsets and rhymes (Laganaro, 2019; Levelt and Wheeldon, 1994). Although this route is likely to be more resource-costly, slower, and more error-prone than the retrieval route, it enables speakers to construct low-frequency or entirely new syllables that are not part of the stored inventory for example, during foreign language learning or when creating new syllables in the native language.

Evidence for the retrieval of stored, precompiled syllables comes from a number of studies showing that high-frequency syllables are produced faster (Bürki et al., 2015; Cholin et al., 2006; Laganaro and Alario, 2006) and with greater accuracy (Tremblay et al., 2016) than low-frequency syllables. These findings seem to be language independent, as effects have been found in a number of different languages such as Dutch (Cholin et al., 2006; Levelt and Wheeldon, 1994), English (Cholin et al., 2011; Croot et al., 2017), French (Laganaro and Alario, 2006), Spanish (Carreiras and Perea, 2004) and Korean (Simpson and Kang, 2004). Studies that systematically compared immediate vs. delayed naming of high- and low-frequency syllables suggest that syllable frequency effects occur at the level of phonetic encoding (Cholin and Levelt, 2009; Croot et al., 2017; Laganaro and Alario, 2006).

Importantly, it is still unclear, whether there is a strict division of labour between the two hypothesized routes within the dual-route account of phonetic encoding, or whether the retrieval and the assembly route always run in parallel, with the output being delivered by whichever route is fastest. The dual-route assumption is supported by two studies by Bürki et al. (2015, 2020), which examined both event-related potentials (ERPs) and reaction times in disyllabic pseudoword production in which the initial syllables were either high-frequency, very low-frequency, or novel syllables. Novel syllables were found to have slower reaction times and more positive ERP amplitudes than high-frequency syllables. Furthermore, the ERPs showed different topographies (around 170ms before speech onset) for high-frequency and novel syllables. However, the syllable frequency effects disappeared when novel syllables were trained, suggesting that novel syllables can be updated to high-frequency syllables, possibly with at least a temporary storage in the syllabary (Bürki et al., 2020).

FMRI studies investigating syllable frequency effects have reported a decrease in activity in regions associated with phonetic encoding when syllable frequency was systematically increased (Carreiras et al., 2006; Papoutsi et al., 2009; Tremblay et al., 2016). These findings of sensitivity of brain activity to qualitatively different mechanisms support the dual-route account that proposes that the retrieval of stored, ready-made high-frequency syllables outruns (the assembly of) low-frequency syllables. However, it is controversial whether the two routes can be attributed to different neurological networks as not all fMRI studies found an effect for high frequency syllables (Brendel et al., 2011; Papoutsi et al., 2009; Riecker et al., 2008).

1.4. The current study

Given the considerable evidence linking stuttering to phonological/phonetic encoding levels, an investigation of syllable frequency effects in AWS seems logical. We hypothesize that AWS benefit even more than fluent speakers from the retrieval of stored syllable units, and that the assembly of low-frequency syllables may be even more costly in an already fragile system, leading to more dysfluencies and more phonetic distortions. We therefore predict that syllable frequency effects may be even more pronounced in AWS than in fluent speakers.

We present high- and low-frequency syllables as first syllables in pseudowords of differing length, i.e., two-, three and four-syllable pseudowords. Almost all of the cited studies above used pseudowords when testing for syllable frequency and found such effects most reliably on frequency manipulated first syllables as opposed to frequency manipulated later syllables. The use of pseudowords allows for a systematic manipulation of syllable frequency independent of factors such as word frequency and other potential confounds such as phonetic complexity of the syllables and sub-syllabic frequencies (segmental and bigram frequencies).

In addition, multisyllabic pseudowords also provide an opportunity to investigate effects of word length, which have been found in previous studies to result in fewer speech errors (e.g. Sasisekaran and Weisberg, 2014) and fewer stuttering events in AWS (e.g. Brown, 1938b, 1945; Logan and Conture, 1995) for shorter versus longer words, which may reflect a comparatively greater encoding effort for longer words.

Effects of syllable frequency and word length have both been located at the interface of phonological and phonetic encoding, i.e., where (i) phonetic syllables are either retrieved from storage or assembled and (ii) where consecutive syllables are chunked into larger metrical frames of multisyllabic words. While high-frequency syllables can be retrieved from storage and may require comparatively less effort than low-frequency syllables that need to be assembled, shorter two-syllable pseudowords require less integration than three- and four-syllable pseudowords.

To summarize, in the current study, we investigated whether AWS' productions of multisyllabic pseudowords are sensitive to manipulations of syllable frequency and pseudoword length: It was expected that AWS would show higher stuttering rates and more errors when producing low-frequency syllables as initial parts of pseudowords compared to when producing high-frequency syllables as initial parts of pseudowords. We also predicted an effect of pseudoword length: the production of longer pseudowords (pseudowords with three or four syllables) was expected to result in higher stuttering rates and more errors than the production of shorter pseudowords (pseudowords with two syllables).

2. Methods

2.1. Participants

The ethical review board of the University Medical Centre Göttingen, Georg August University Göttingen, Germany, approved the study. All participants provided written informed consent, according to the Declaration of Helsinki, before any study-related procedure took place. Recruitment of participants was accomplished via advertisements on University's black boards and on events of the German stuttering self-help organization (BVSS). Twenty participants per group, i.e., 20 adults with developmental stuttering (AWS) and 20 matched fluently speaking control participants (FC) were tested. Most AWS had received stuttering interventions, but were asked not to use any speech technique in the current study. Exclusion criteria included any speech or language disorder other than developmental stuttering, neurological impairment, drug abuse, or medications that act on the CNS. We excluded the data of one participant due to an incomplete dataset and another participant because of a medicated depression that was reported only after the experiment was finished. Thus, we analysed 19 AWS (4 females, mean age 27.8 ± 10.5), and 19 fluent speakers (3 females, mean age 26.3 ± 8.3). The groups were comparable with regard to age, sex, years of education and handedness (Oldfield, 1971) (Table 1). Speech fluency was assessed by A.K. (a certified clinical linguist) with the stuttering severity index (SSI-4; Riley, 2009) prior to the experiment on the basis of 300 words of spontaneous speech and 300 words of reading. In the stuttering group, four participants were diagnosed with very mild stuttering, four with mild stuttering, and six with more moderate stuttering, while two participants stuttered severely, and three participants very severely. Inter-rater reliability was established by reanalysing nine random participants from each group by a second speech and language pathologist, N.E.N. Intraclass correlations of the SSI-4 scores yielded a good to excellent inter-rater reliability (total: ICC = 0.95, 95 % CI [0.87, 0.98]; spontaneous: ICC = 0.95, 95 % CI [0.75, 0.99]; reading: ICC = 0.92, 95 % CI [0.75, 0.98]; duration: ICC = 0.83, 95 % CI [0.59; 0.93]; concomitants: ICC = 0.78, 95 % CI [0.48, 0.91]). To establish group

Table 1
Demographic information of participants.

	AWS	FC	Test-statistics (df)	2-sided p-value
n	19	19		
Age, years	27.8 ± 10.5	26.3 ± 8.3	183 [i]	0.953
Sex ratio	15:4	16:3	– [ii]	1
Education, ranks[a]	3 (1)	3 (0)	215.5 [i]	0.219
Handedness, laterality quotient	88.9 (15.8)	87.5 (29.8)	149 [i]	0.353
SSI-4	25 (12.5)	6 (3.5)	–	–
OASES	2.1 ± 0.7	–	–	–
Onset, years	4.8 ± 3.0	–	–	–
Well-being index	18 (5)	16 (7.5)	139 [i]	0.229
BDI	2 (4.5)	3 (8)	215.5 [i]	0.309
SLRT – II words, n	104.1 ± 20.1	114.2 ± 19.9	3.13 (1,35)[iii]	0.466
SLRT – II pseudowords, n	64.5 ± 13.04	77.3 ± 14.1	0.21 (1, 35)[iii]	0.652
Konsonanten Trigram Test, n	45.6 ± 8.4	48.6 ± 5.6	1.32 (31.29)[iii]	0.196
Digit span f	9.7 ± 1.7	10.3 ± 1.9	3.13 (1,35) [iii]	0.085
Digit span b	8.6 ± 1.6	10.1 ± 1.9	2.00 (1,35)[iii]	0.166

Note. Interval/ratio -scaled variables are presented as mean ± standard deviation. Ordinal-scaled variables are presented as median (interquartile range). [i]Mann-Whitney-U test, [ii]Fisher's exact test, [iii]ANCOVA with stuttering severity (SSI-4 total score) as confounding covariate.

[a] 1 = high school, 2 = <2years college, 3 = 2 years of college, 4 = 4 years of college. AWS: adults who stutter; FC: fluently speaking control participants.

comparability and to ensure that participants had no clinically significant deficits in cognitive functions possibly influencing reading behaviour, seven additional subtests were administered. The WHO test of mental well-being (World Health Organization, 1998) and the Beck depression inventory (version II, Beck et al., 1996) were used for assessing mental well-being. Working memory was tested with the digit span forwards and backwards subtests of the German version of the Wechsler adult intelligence scale – fourth edition (Petermann, 2012) and the Konsonanten Trigram Test (Consonant Trigram Test, Schellig & Schächtele, 2002). Reading performance was assessed with the 1-min word and pseudoword reading tests of the Salzburger Lese- und Rechtschreib Test – II (SLRT – II; Moll and Landerl, 2010). Both the Digit span and the SLRT - II required the repetition of items with increasing sequence length; we therefore included stuttering severity as confounding covariate. Groups were comparable in mental well-being, reading performance, working memory and short-term memory (Table 1).

In addition, we used a self-assessment of the psycho-social impact of stuttering (Overall Assessment of the Speaker's Experience of Stuttering, OASES) (Yaruss et al., 2016). Detailed information of assessments is provided in Supplementary Table 1.

2.2. Materials

The material set of 300 pseudowords was constructed on the basis of 50 high-frequency and 50 low-frequency German syllables. High- and low-frequency syllables were paired (e.g.,/pat/HF,/po:t/LF) to serve as initial syllables in two-, three- and four-syllabic pseudoword pairs. The pseudoword pairs of different length only differed by their initial syllable, i.e., the appended second (in two-syllabic pseudowords), second and third (in three-syllabic pseudowords) and second, third and fourth syllables were identical within pseudoword pairs. For example, the pseudowords constructed based on the initial high-frequency syllable/ pi:/were/pi:rɪʃ/,/pi:rɪʃbe:/,/pi:rɪʃbe:o:l/; the low-frequency counterparts were/pɔɪrɪʃ/,/pɔɪrɪʃbe:/,/pɔɪrɪʃbe:o:l/. In this way, the 100 two-, 100 three-, and 100 four-syllabic pseudoword pairs were controlled for a number of phonetic factors (see below). The 300 pseudowords were formed by following phonological, phonetic and orthographic rules of the German language. Care was taken that none of the syllables or the multisyllabic pseudowords constituted lexicalized forms. A full list of items and their orthography can be found in the Supplementary Table 2.

Low-frequency syllables had an occurrence of less than 110 times per million (mean: 11, median: 0.5, range: 0–98), whereas syllables with a token frequency above 110 occurrences per million (mean: 2114, median: 780, range = 117–16467) were categorized as high-frequency syllables. The summed token syllables (summed frequency of occurrence of individual syllables) were drawn from Hofmann et al. (2007) who used the phonological word form lexicon of the CELEX database (CEntre for LEXical Information; Baayen et al., 1996). To reduce the influence of linguistic factors other than syllable frequency type, high- and low frequency initial syllables of each pseudoword pair resembled each other phonetically except for the syllable nucleus (e.g.,/pi:/(HF) vs./pɔy/(LF)). Therefore, the phonetic complexity of high- and low-frequency syllables was equal. All pseudowords began with a bilabial or labio-dental consonant (/p/,/b/,/f/,/v/,/m/). Initial syllables had a CV (n = 36), CVC (n = 210), CCV (n = 48) or CCVC (n = 6) structure.

As AWS are known to stutter less in experimental settings than during daily situations (Jackson et al., 2020), it is necessary to administer many trials and/or experimental stimuli with high phonetic complexity (Dworzynski and Howell, 2004) in order to be able to elicit enough stuttering events. As previous studies had difficulties to generate enough stuttered trials (Mersov et al., 2018; Vanhoutte et al., 2016), we used 300 trials of overt speech production in the current study. We also ensured high phonetic complexity of stimuli by adhering to the index of phonetic complexity (IPC) (Dworzynski and Howell, 2004) for German. According to these authors, German speaking AWS show a higher risk to stutter if content words contain dorsal sounds (IPC factor 1), fricative, affricative or liquid sounds (IPC factor 2), if they end with a consonant (IPC factor 5), if they have more than three syllables (IPC factor 6) or if they contain consonant clusters (IPC score 7) with heterorganic articulation places (IPC score 8). Thus, all pseudowords had to score in at least four out of the six mentioned factors. Moreover, segment length (number of letters and phonemes) as well as orthographic and phonemic bigram frequency were counterbalanced (Table 2). Pseudowords grouped by frequency type of the initial syllable did not differ in phonetic complexity (ICP) or segment length. Initial syllables grouped by syllable frequency were comparable in the frequency of the first and second orthographic bigram, but differed in frequency of the third orthographic and all phonetic bigrams (Table 2). The nucleus of initial syllables grouped by syllable frequency differed for vowel length but not for place of articulation (Table 2). In the experimental set, the 300 speech stimuli were duplicated, to serve as items in the overt as well as in the covert reading condition.

2.3. Design

We used a 2 × 2 × 3 design with the factors group (AWS vs. FC), syllable frequency (high- vs. low-frequency initial syllables) and pseudoword length (two-syllabic, three-syllabic, four-syllabic pseudowords). Group is the only between participants factor, all other factors were tested within participants. Syllable frequency was tested between items; reading condition and pseudoword length were tested within items. The 600 pseudowords were divided into eight blocks consisting of 75 items

Table 2
Linguistic factors of pseudowords and initial syllables.

Unit	Linguistic factors	Descriptives of HF/LF	Test statistics
Pseudoword	IPC Score	7 (3)/7 (3)	[a]
	letters	9 (4)/10 (4)	$W = 10608$, $p = .389$, $r = -0.05$
	phonemes	8 (4)/8 (4)	$W = 11253$, $p = .997$, $r < -0.01$
Initial syllable	1st bigram, orthographic	23,447 (54,780)/ 20,132 (40,019)	$W = 1139$, $p = .853$, $r = -0.01$
	2nd bigram, orthographic	46,490 (105,267)/ 21,016 (68,432)	$W = 12474$, $p = .008$, $r = -0.16$
	3rd bigram, orthographic	20,0537 (432,552)/ 45,650 (139,543)	$W = 2754$, $p < .001$, $r = -0.39$
	1st bigram, phonemic	23,357 (47,100)/ 13,706 (20,122)	$W = 14121$, $p < .001$, $r = -0.22$
	2nd bigram, phonemic[b]	35,255 (66,782)/ 10,178 (26,992)	$W = 12794$, $p < .001$, $r = -0.32$
	CV bigram[c], phonemic	26,045 (59,027)/ 9612 (21,524)	$W = 15048$, $p < .001$, $r = -0.29$
Nucleus initial syllable	vowel length	long vowel: 60/ 105 short vowel: 90/ 45	*Chi^2 (1) = 7.92, p = .005, odds ratio: short vowels occur 3.5 times more often in HF than in LF*
	place of articulation	anterior: 117/96 posterior: 33/54	*Chi^2 (1) = 6.48, p = .011, odds ratio: posterior vowels occur 2.0 more often in LF than in HF syllables*

Note. Descriptives are given as median (IQR) or as total number.
[a] Please note that pseudowords grouped by frequency type had exactly the same IPC scores.
[b] Bigram frequency of phonemes/letters occurring in 3rd and/or 4th position are not reported as only few pseudowords had initial syllables with more than 4 phonemes/letters (n < 20). Significance level: $p = .005$ [Bonferroni corrected for multiple comparisons 0.05/10].
[c] CV bigram represents the 1st bigram for CV or CVC syllables and the 2nd bigram for CCV and CCVC syllables.

each. All blocks contained an even number (±1) of overt and covert reading[1] conditions, high- and low-frequency pseudowords, as well as two-, three-, and four-syllabic pseudowords. Within blocks, reading condition, syllable frequency and pseudoword length were counterbalanced such that after maximal five items of each condition an intervening item was presented. Regarding the factor reading condition, within each block, pseudowords appeared always first in the overt condition before being presented in the covert condition. With regard to syllable frequency, either the high- or the low-frequency pseudoword of one pair (i.e.,/pi:rɪʃ/or/pɔɪrɪʃ/) was presented. Pseudowords of different length but with the same initial syllable never occurred in the same block. Thus, each block contained an even (±1) number of two-, three and four-syllabic pseudowords with different initial syllables in a pseudorandomized order. In addition, for half of the participants within each group, all pseudowords were replaced by their syllable frequency counterpart, reversing the sequence of syllable frequency. The order of blocks was counterbalanced within and across groups following a latin-square design.

2.4. Procedure

Participants were tested individually in a quiet room. During the experiment, participants sat in a comfortable chair at 90 cm distance to the computer screen wearing an EEG high-density net. The task instruction and stimuli were presented visually on the centre of the screen using the experimental software PsychoPy, version 1.85.2 (Peirce, 2008; Peirce et al., 2018).

Participants were instructed to read the visually presented pseudowords aloud when a black cross preceded them but to read the pseudowords silently when a white cross preceded them. The stimuli for task implementation of the reading condition (i.e., cross) were presented 2 s before the appearance of the pseudowords. In addition, an inter-stimulus interval of 2.5–3.5 s after the presentation of the pseudoword and before the next task implementation separated trials. Pseudowords in the overt reading condition were presented for 6 s, whereas the presentation of pseudowords was reduced to 2 s in the covert condition. After each block a planned pause, which allowed participants for positional readjustment or to drink was implemented. Every participant followed the same sequence of pause durations and test assessments.

As a self-initiated speech preparation was used, participants were instructed to wait for an internal urge to speak before overtly reading the presented pseudoword (McArdle et al., 2009). Because prosodic unknown pseudowords were used, participants were asked to pretend as if reading a German word. In addition, participants were instructed not to use any fluency enhancing technique, not to suppress stuttering events or to correct themselves. Ten practice trials, in which participants were informed on their response correctness, were conducted before starting the experiment.

2.5. Data preprocessing

During the experiment, participants were videotaped. Two speech and language pathologists, K.H. and J.W., transcribed and evaluated speech samples of all participants for stuttering events, errors and their syllable position using ELAN, version 5.8 (Max Planck Institute for Psycholinguistics, 2018). Inter-rater reliability was maintained by comparing the evaluation of four randomly chosen participants (two per group). Cohen's Kappa was calculated to estimate inter-rater reliability, considering that the data are binary and exhibit an imbalanced distribution. Stuttering events and response accuracy were identified with inter-rater reliabilities of 0.80 and 0.75, respectively. Kappa values between 0.61 and 0.80 fall within the range of substantial agreement (Landis and Koch, 1977). In addition, the first author (AK) checked all transcriptions for consistency. Some orthographic pseudowords allowed different phonetic implementations, as participants did not listen to a correct pronunciation of the pseudowords. Ambiguous realizations of a pseudoword included incorrect syllabification (e.g.,/ge: us/instead of/gɔys/) and incorrect vowel length of initial syllables (e.g.,/vu: k/instead of/vʊk/). Such realizations were marked as 'unclear' errors. Pseudowords with unclear errors as well as trials with missing verbal responses were excluded from further statistical analysis. To keep the stimuli material balanced, the categorical frequency partners of these discarded pseudowords were excluded as well.

2.6. Dependent variables

Two dependent variables were measured: (1) speech fluency, i.e., stuttering events and (2) response accuracy, i.e., errors. Responses were either marked for being correct or for containing a stuttering event or an error.

Stuttering events in AWS were registered using the SSI-criteria (Riley, 2009). These consist of sound and syllable repetitions or prolongations as well as blocks with visible speech effort. Sound and syllable repetitions that were preceded by an error were not counted as stuttering symptoms but as self-corrections.

Errors in both groups were defined as additions, elisions, substitutions and metathesis of consonants and vowels. As in Tremblay et al. (2016), substitutions of vowels in medial or final syllables addressing their tensity or length (e.g.,/o/vs./ɔ/,/u/vs./ʊ/or/a/vs./a:/) were not considered as errors as these might be viewed as correct adaptation due to a change in word stress.

Multiple occurrences of one single variable within the same pseudoword were counted as one event (Sasisekaran and Weathers, 2019; Yaruss, 1999). We excluded pseudowords and their pairs containing an error and stuttering event (AWS: 2.12 %, FC: 0.34 %) from statistical analyses (Coalson and Byrd, 2017).

2.7. Statistical analyses

We calculated two main and two additional Generalized Linear Mixed Models (GLMM, Baayen et al., 2008), with binomial error structure and logit link function. The main GLMMs investigated the influence of syllable frequency on stuttering events or errors over all syllable positions. The additional GLMMs tested only stuttering events or errors that occurred at the initial syllable position.

The final full models are described in the result section; here the procedure of building the models is presented. GLMMs included the experimental factors and their interactions as within- or between-subjects fixed effects. To account for the individual level of speech fluency performance in the reading task, we included stuttering severity (summed SSI - IV scores of reading and spontaneous speech) to the models of stuttering probability and reading fluency performance (number of read pseudowords, SLRT-II) to the models of error probability as z-transformed covariates. Several linguistic factors revealed differences between pseudoword pairs (see Table 2). To check whether these factors influenced the variance of our data, we added three variables separately as fixed effects to the full models: first phonemic bigram frequency, CV phonemic bigram frequency and vowel length (for more details see Table 2). Second phonemic bigram frequency was not added, because not all pseudowords contained a second bigram within their initial syllable. Factors that significantly improved the model as indicated by likelihood ratio tests (R function *anova* with argument test set to "Chisq") were kept (Supplementary Table 3).

We included all possible random effects and slopes, to keep type I error at the nominal level of 5 % (Barr et al., 2013). Possible random terms were determined by following the procedure of Bates et al. (2018). We first fitted the maximal model including all possible random effect components and then iteratively reduced the model from random terms

[1] The covert reading condition served as control task of the larger EEG study (Korzeczek et al. (2022)).

that were not supported by the data. Reduction steps were based on principal component analyses using the function *rePCA* and likelihood ratio tests (Bates et al., 2018).

Model assumptions were checked by inspecting visually the distribution of residuals, by assessing model stability (Supplementary Tables 4 and 5) and by ruling out collinearity with the variance inflation factors being beyond 1.5 (Field et al., 2013). The variance inflation factor was derived by the function *vif* of the R-package car (Fox and Weisberg, 2019), applied to a standard linear model excluding the random effects. All models met the assumption criteria.

As an overall test of the effect of the fixed effects and their interactions we compared the full model with a null model lacking the fixed effects (Field et al., 2013) using a likelihood ratio test. The statistical significance of fixed effects was tested based on likelihood ratio tests, comparing the full model with the respective reduced models (R function *drop1*). Reduced models comprised the same random effects structure as the full model. The models were fitted in R (version 3.6.2, R Core Team, 2019) using the function *glmer* of the R package lme4 (version 1.1–21) (Bates et al., 2015). Confidence intervals were derived using the function *bootMer* of the package lme4 with 1000 parametric bootstraps. To explain effect sizes of individual fixed effects, odd ratios (OR) were plotted using the function plot_model of the package sjPlot (Version 2.8.1) (Lüdecke, 2013). For example, an OR of 2 for the fixed effect syllable frequency would mean that the odds are twice as high for high-frequency syllables compared to low-frequency syllables, indicating a moderate effect. Respectively an odds ratio of 0.5 would mean that the odds for high-frequency syllables are half as high as for low-frequency syllables. As a rule of thumb, odd ratios of around 1.5 could be considered as a smaller effect, whereas odd ratios larger than 3 could be considered as large effect (Sullivan and Feinn, 2012). To interpret an odds ratio (OR) less than 1, we took the reciprocal by dividing 1 by the OR.

2.8. Premise for subsequent analyses: groups are comparable with regard to unclear errors

To ensure that the discarded unclear errors (for a definition see 2.5) did not influence our analyses on syllable frequency effects, we run a factorial two-way ANOVA with Group as between-subjects factor and syllable frequency as within-subjects factor. On average, AWS produced 29.32 (SD = 12.12) unclear errors in high-frequency and 31.16 (SD = 12.06) unclear errors in low-frequency pseudowords. FC produced on average 25.11 (SD = 6.44) unclear errors in high-frequency and 27.95 (SD = 5.73) unclear errors in low-frequency pseudowords. Results showed no main effect of group, $F(1,72) = 2.86$, $p = .09$, $\omega^2 = 0.92$, no effect of syllable frequency, $F(1,72) = 1.14$, $p = .29$, $\omega^2 = 0.25$, and no interaction, $F(1,72) = 0.05$, $p = .82$, $\omega^2 = -1.17$. Thus, groups did not differ by unclear reading errors, nor did unclear errors affect a specific frequency category. For statistical hypotheses testing 7570 (FC: 3958; AWS: 3612) pseudowords remained.

3. Results

3.1. Pseudoword length but not syllable frequency influences stuttering probability when considering all syllable positions

The first generalized linear mixed-effects model involved stuttering events as the dependent variable and syllable frequency (high- and low-frequency of initial syllables), the z-transformed covariate of pseudoword length and stuttering severity and the two-way interaction of pseudoword length and syllable frequency as independent variables. Overall, 2772 items were included in the analysis (for descriptive statistics see Table 3).

The model was significant compared to the null model (χ^2 (4) = 98.03, p < .001). Stuttering frequency during pseudoword reading was significantly influenced by pseudoword length (χ^2 (1) = 82.81, $p < .001$, β = 0.53, SE = 0.06, 95 % CI [0.41, 0.65]). Hence, the longer the pseudoword the more likely the occurrence of stuttering (Fig. 1A and B). In addition, stuttering severity predicted the number of stuttering events (χ^2 (1) = 11.8, $p < .001$, β = 1.31, SE = 0.33, 95 % CI [0.66, 1.97]). The more severe the stuttering of a participant, the more likely it was that stuttering events occurred during the experiment (Fig. 1A–C). The initial syllable frequency showed a trend-level significance (χ^2 (1) = 3.39, p = .06), which falls below the threshold for a small effect size. The interaction between syllable frequency and pseudoword length was not significant (χ^2 (1) = 0,33, p = .566). (Fig. 1A).

Table 3
Frequency of stuttered events over all syllable positions in AWS for two-, three- and four-syllabic pseudowords.

Stuttered events	two- syllabic		three- syllabic		four- syllabic	
	HF	*LF*	*HF*	*LF*	*HF*	*LF*
total responses (n)	526	526	471	471	389	389
stuttered events (n)	85	74	112	96	118	114
stuttered events (%)	16.2	14.1	23.8	20.4	30.3	29.3
stuttered events (SD in %)	36.8	34.8	42.6	40.3	46.0	45.6

3.2. Trend of syllable frequency effect on stuttering probability in first syllable position

Next, we investigated the effect of syllable frequency on the probability to stutter on syllables in initial word positions. The model consisted of stuttering events as the dependent variable as well as syllable frequency (high- and low-frequency of initial syllables), pseudoword length, stuttering severity, and the two-way interaction of pseudoword length and syllable frequency as independent variables. In addition to these fixed effects, vowel length, which contributed significantly to the model (Supplementary Table 3), was added to the full model. Overall, 2458 pseudowords were included in the analysis (for descriptive statistics see Table 4).

The model was significant compared to the null model (χ^2 (5) = 30.5, $p < .001$). Again, the occurrence of stuttered events was significantly influenced by pseudoword length (χ^2 (1) = 13.39, $p < .001$, β = 0.28, SE = 0.08, 95 % CI [0.12, 0.44]) and stuttering severity (χ^2 (1) = 16.13, $p <$ .001, β = 1.92, SE = 0.42, 95 % CI [1.09, 2.93]) (Fig. 2A). Neither syllable frequency nor vowel length were statistically significant (syllable frequency: χ^2 (1) = 0.58, p = .445, vowel length: χ^2 (1) = 1.11, p = .292). There was a trend in the opposite direction: 12 out of 19 (63 %) of our participants stuttered more on high- than on low-frequency first syllables (Fig. 2B–Supplementary Table 6).

3.3. Pseudoword length but not syllable frequency influences response accuracy when considering all syllable positions

For modelling errors, we included syllable frequency (high- and low-frequency initial syllables), group (AWS and FC) and the z transformed covariates of pseudoword length and reading fluency performance (SLRT-II pseudoword score), the three-way interaction between group, frequency and length and their underlying two-way interactions. Fixed effects of vowel length or token frequency of phonemic bigrams did not improve the model, all $p > .05$. Overall, 6170 items were included in the analysis (for descriptive statistics see Table 5).

The model was significant compared to the null model (χ^2 (8) = 47.59, $p < .001$). The occurrence of errors was significantly influenced by pseudoword length (χ^2 (1) = 35.99, $p < .001$, β = 0.55, SE = 0.07, 95 % CI [0.4, 0.69]). Participants produced more errors on three- and four-syllabic than on two-syllabic pseudowords (Fig. 3A and B). In addition, reading performance predicted the occurrence of errors (χ^2 (1) = 8.31, p = .004, β = −0.31, SE = 0.1, 95 % CI [−0.52, −0.11]), indicating that participants with low score in the SLRT-II pseudoword task made more errors (Fig. 3A–C). Neither initial syllable frequency nor group or the

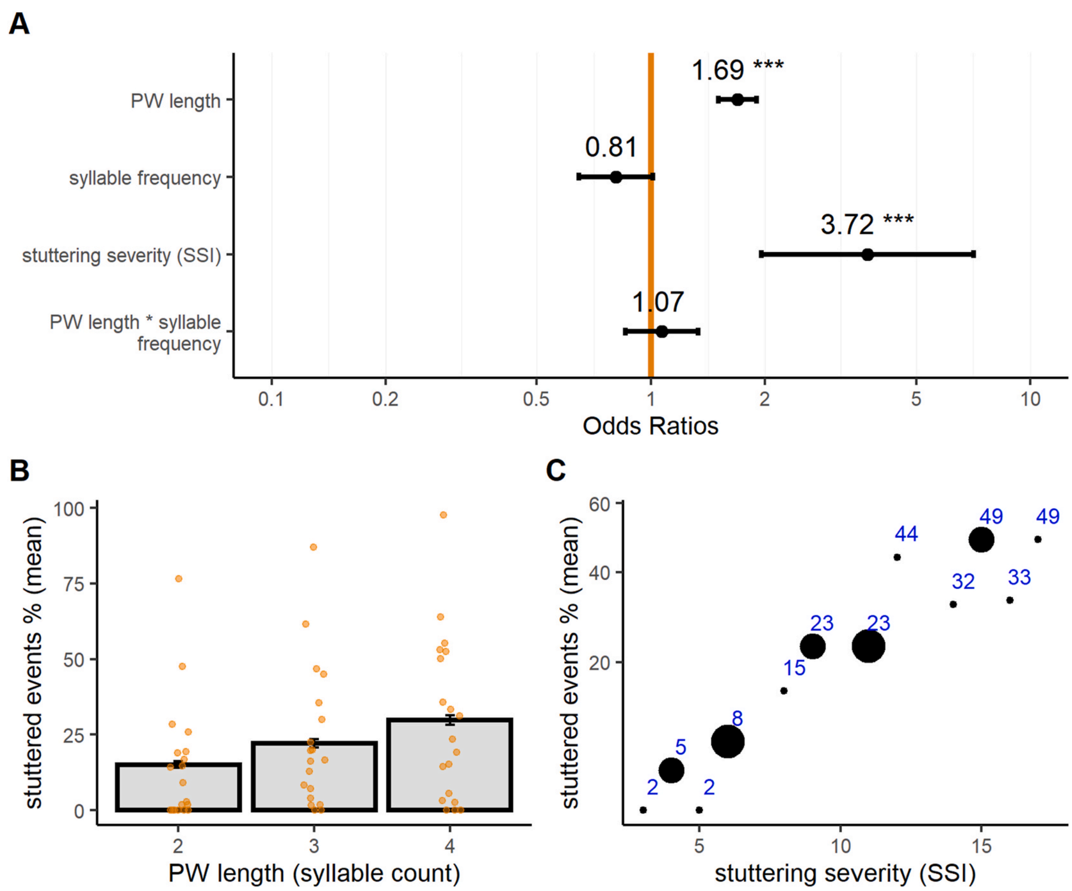

Fig. 1. Main effects of pseudoword length and syllable frequency on stuttering occurrence. Note. A. Stuttering probability as indicated by Odds ratios was significantly influenced by pseudoword (PW) length (increase in syllables) and stuttering severity. Whiskers indicate confidence intervals at 0.95. Contrasts of syllable frequency were set to −0.5 (LF) vs. 0.5 (HF). B. Main effect of pseudoword length: Bar plots, represent means of stuttered events for two-, three- and four-syllabic pseudowords. Points represent individual means of participants. Error bars represent standard errors. C. Main effect of stuttering severity: Scatterplot depicts mean of stuttered events according to stuttering severity, point size depicts the number of participants diagnosed with the respective stuttering severity score (small point: 1 participant, largest point: 3 participants).

Table 4
Frequency of stuttered events in initial syllable positions in 19 AWS for two-, three-, and four-syllabic pseudowords.

Stuttered events	two- syllabic		three- syllabic		four- syllabic	
	HF	*LF*	*HF*	*LF*	*HF*	*LF*
total responses (n)	510	510	410	410	309	309
stuttered events (n)	76	62	70	53	57	56
stuttered events (%)	14.9	12.2	17.1	12.9	18.4	18.1
stuttered events (SD in %)	35.6	32.7	37.7	33.6	38.8	38.6

two- and three-way interactions were statistically significant (Fig. 3A).

3.4. AWS are less accurate on high-frequency initial syllables in two- and three-syllabic pseudowords than FC

For modelling errors in initial syllable position as response variable, we included syllable frequency, group and the z-transformed covariates of pseudoword length and reading performance and all possible interactions between group, frequency and length. Vowel length or token frequency of phonemic bigrams did not improve the model, all $p > .05$. Overall, 5404 items were included in the analysis (for descriptive statistics see Table 6).

The model was significant compared to the null model (χ^2 (8) = 30.12, $p < .001$). Response errors were influenced by the interaction between syllable frequency and group (χ^2 (1) = 10.02, p = .002, β = −1.38, *SE* = 0.45, 95 % CI [−2.44, −0.59]). Contrary to our hypothesis, AWS made more errors on pseudowords with initial high-frequency syllables compared to pseudowords with initial low-frequency syllables, whereas FC showed the opposite pattern (Fig. 4A and B). Furthermore, we found an interaction between initial syllable frequency and pseudoword length with (χ^2 (1) = 4.5, p = .002, β = 0.34, *SE* = 0.16, 95 % CI [0.03, 0.66]) (Fig. 4A–C). Fig. 4C shows that contrary to two- and three-syllabic words, low-frequency four-syllabic pseudowords elicited considerably more errors than high-frequency four-syllabic pseudowords. This statistically significant two-way interaction should be regarded with caution. Although, the three-way interaction did not become significant, the higher error rate for high-frequency two-syllabic and three-syllabic pseudowords compared to their low-frequency counterparts was only evident in AWS (Table 6). Again, pseudoword length (χ^2 (1) = 6.49, p = .011, β = 0.21, *SE* = 0.08, 95 % CI [0.04, 0.37]) and reading performance (χ^2 (1) = 7.96, p = .005, β = −0.46, *SE* = 0.15, 95 % CI [−0.78, - 0.15]) predicted the occurrence of errors.

4. Discussion

This study investigated whether stuttering is associated with planning difficulties at the phonetic-phonological interface. Given the syllable frequency effects observed in fluent speakers, we expected high-frequency syllables to be produced more fluently and more accurately than low-frequency syllables also by our non-fluent participants. When analysing stuttering events, there was a trend in the opposite direction.

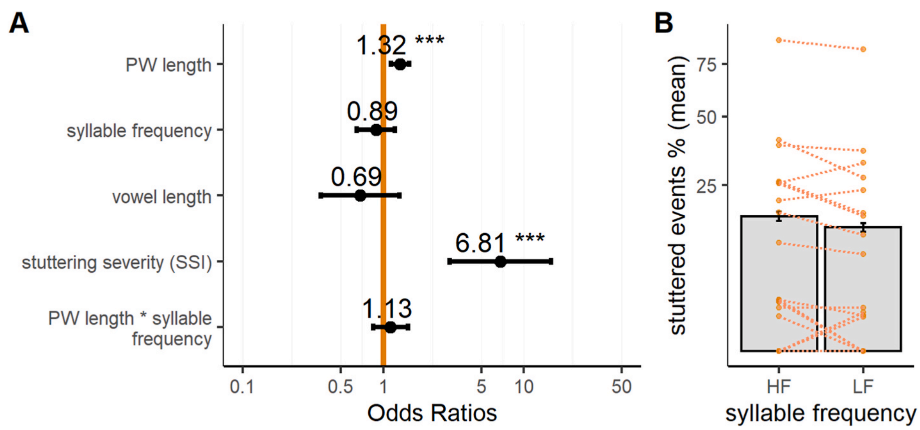

Fig. 2. Probability of stuttered events at initial syllable position. Note. A. Stuttering probability as indicated by Odds ratios was significantly influenced by pseudoword (PW) length (increase in syllables) and stuttering severity. Contrasts of syllable frequency were set to −0.5 (LF) vs. 0.5 (HF). B. Bar plots represent means of stuttered events for high- and low-frequency pseudowords. Points represent individual means of participants. Error bars represent standard errors.

Table 5
Frequency of errors at all syllable positions per group for two-, three-, and four-syllabic pseudowords ($n_{AWS} = 19$, $n_{FC} = 19$).

Group	Errors	two- syllabic		three- syllabic		four- syllabic	
		HF	LF	HF	LF	HF	LF
FC	total responses (n)	664	664	634	634	579	579
	errors (n)	36	39	73	63	89	105
	errors (%)	5.4	5.9	11.5	9.9	15.4	18.1
	errors (SD in %)	22.7	23.5	31.9	29.9	36.1	38.6
AWS	total responses (n)	471	471	405	405	332	332
	errors (n)	34	33	47	44	53	67
	errors (%)	7.2	7.0	11.6	10.9	16.0	20.2
	errors (SD in %)	25.9	25.6	32.1	31.2	36.7	40. 2

This inverse syllable frequency effect was statistically significant in response accuracy of the initial first syllable of pseudowords: AWS produced *more* errors on pseudowords containing high-frequency first syllables compared to those pseudowords with low-frequency first syllables. Fluently speaking control participants showed the opposite non-significant trend. AWS' tendency to produce more errors on high-frequency syllables compared to low-frequency syllables included another interesting finding: in *short* pseudowords, i.e., two- and three-syllabic, more errors were found in words with high-frequency first syllables compared to their low-frequency counterparts. However, this pattern was reversed for *long* pseudowords. In addition, the main effect of pseudoword length showed a robust effect, confirming our hypothesis that the number of stuttering events and response errors would increase with pseudoword length. In the following sections, we discuss psycholinguistic mechanisms that could produce the inverted syllable frequency effect and that could also explain word length effects in AWS. We also discuss the results in the light of neural evidence and neurocomputational modelling.

4.1. Interaction of syllable frequency and group

The dual-route account offers an explanation for the advantage of high-frequency syllables over low-frequency syllables in fluent speakers by assuming that the retrieval of precompiled units is faster and less error-prone compared to a segment-by-segment construction. The foundation for a store of highly trained motor programs is laid in the early stages of speech acquisition, when syllables are learned by imitating and repeating speech sounds, which are repeatedly adapted via sensorimotor links (Levelt et al., 1999; see also Kearney and Guenther, 2019). In a fragile system such as in DS, we expected more pronounced syllable frequency effects based on the hypothesis that the reliance on early acquired, high-frequency syllables would be even greater and the assembly process, thought to be more resource demanding, would be more laborious and error-prone in AWS. However, the reverse pattern was observed in the results reported above. One possible explanation for these unexpected results is the assumption that the dual-route mechanism operates differently in AWS. With respect to the retrieval route, AWS may not acquire and store motor programs in the same way as fluent speakers do, resulting in more fragile motor programs and, as a consequence, to a lower efficiency in phonetic encoding. In stuttering research, there is still an ongoing debate about the evidence for impaired sequence learning or automaticity in AWS in both speech and non-speech, i.e., finger tapping, tasks (e.g. Höbler et al., 2022; Smits-Bandstra and Nil, 2009; Smits-Bandstra and Nil, 2013; Smits-Bandstra et al., 2006a, 2006b, but see Korzeczek et al., 2020; Masapollo et al., 2021; Tendera et al., 2020 for similar learning performance in AWS and FC). Nevertheless, a recent study observed a frequency effect in implicit sequence skill performance in developmental stuttering: male AWS showed a better learning performance, i.e., a greater improvement in finger tapping speed for low- compared to high-frequency triplets (Höbler et al., 2022). The idea of inadequately acquired motor programs which may have a number of detrimental effects on speech planning in AWS has not been considered in cognitive accounts of stuttering but may provide an appropriate explanation for the current findings (see Max et al., 2004) for a similar account modeling speech motor control). Within the dual-route account, the following scenarios seem tangible: (i) a program that does not qualify as a stable motor unit takes longer to be selected from a pool of neighbouring syllables and may require more intensive phonetic fine-tuning. (ii) The fragile target syllable is eventually outrun by a neighbouring syllable that competes with the target syllable due the spread of activation within the syllabary network, as the target cannot overcome the activation threshold in time. Thus, a close competitor is eventually selected. This incorrect selection of a stored syllable will lead to conflict with the output delivered by the assembly route. Resolving this conflict will further delay subsequent processing. Either way, only when the syllable boundary, marking the end of the current syllable, has been determined during syllabification procedures, the following syllable can be encoded. Until that point, the processing of subsequent syllables is put on hold, possibly leading to planning delays and blocks. (iii) It may also be the case that the stuttering system, due to persistent problems in accessing motor programs in the syllabary, decides to abandon and bypass the retrieval route altogether in favour of the assembly route instead (for such an account for speech apraxia see Whiteside and Varley (1998); for

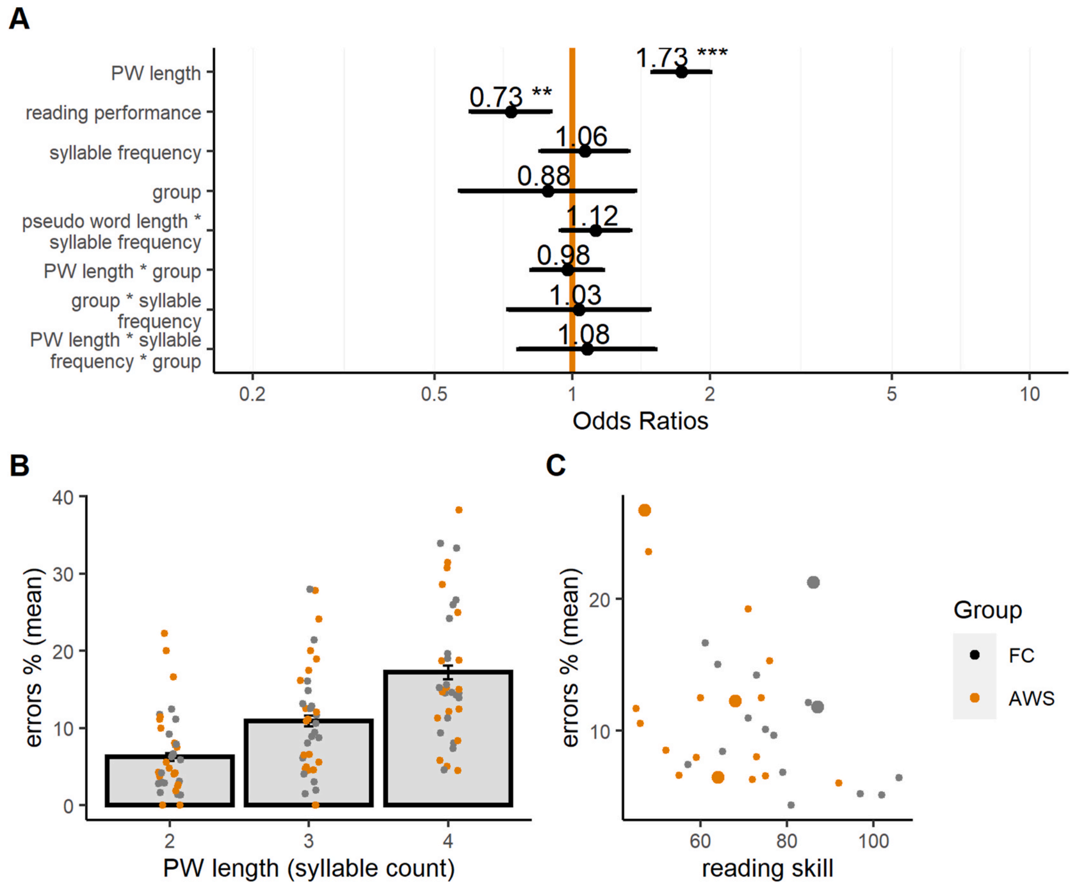

Fig. 3. Probability of errors overall syllable positions ($n_{AWS} = 19$, $n_{FC} = 19$). Note. A. Probability of errors as indicated by Odds ratios was significantly influenced by pseudoword (PW) length and reading skill. B. Main effect of pseudoword length: Bar plots represent Group means of errors for two-, three-, and four-syllabic pseudowords. Points represent individual means of participants. Error bars represent standard errors. C. Main effect of reading skill: Means of errors plotted against score in reading skill (score in SLRT-II pseudoword). Point size depicts the number of participants that received the respective reading skill, small point: one participant, largest point: two participants.

Table 6
Frequency of errors in initial syllable positions per group for two-, three-, and four-syllabic pseudowords ($n_{AWS} = 19$, $n_{FC} = 19$).

Group	Responses in	two- syllabic		three- syllabic		four- syllabic	
		HF	LF	HF	LF	HF	LF
FC	responses (n)	636	636	548	548	466	466
	errors (n)	19	21	13	15	15	24
	errors (%)	3.0	3.3	2.4	2.7	3.2	5.2
	errors (SD in %)	17.0	17.9	15.2	16.3	17.7	22.1
AWS	responses (n)	440	440	352	352	260	260
	errors (n)	18	12	16	10	10	17
	errors (%)	4.1	2.7	4.5	2.8	3.8	6.5
	errors (SD in %)	19.8	16.3	20.9	16.6	19.3	24.8

a neurocomputational account on speech apraxia with damage to syllabic motor programs see Guenther (2016). If the third scenario is true, high-frequency syllables will no longer have an advantage over low-frequency syllables, since now *all* syllables will be built from scratch. Our finding of Group by Syllable Frequency interaction might be more consistent with the assumption of less stable high-frequency syllables, i.e., the first and second scenarios. Taken together, under the hypothesis of fragile motor programs in AWS, all scenarios imply that high-frequency syllables do not have an advantage over low-frequency ones but instead may cause more errors due to incorrectly selected syllables or due to compensatory repair processes. This latter explanation is consistent with the assumption in the DIVA model that fragile motor programs may lead to an overreliance on feedback control in AWS (Civier et al., 2010; Max et al., 2004; Meier and Guenther, 2023). Indeed, when the model parameters in DIVA were adjusted towards a greater reliance on feedback control, the model produced not only more speech errors, but also more segmental and syllable repetitions — one of the primary symptoms of stuttering (Civier et al., 2010).

In addition, the stronger interconnectedness of the individual motor subparts within high-frequency syllables, which facilitates production in fluent speakers could have a detrimental effect on AWS speech, forcing larger motor chunks onto a more fragile system that actually prefers smaller chunks (for arguments regarding insufficient motor control in AWS see Namasivayam and van Lieshout, 2011; Smith and Weber, 2017). It should be noted, however, that sometimes, the retrieval of precompiled larger motor chunks, albeit fragile or only loosely connected, may occasionally be judged as ’good‘ enough to guide the subsequent processes towards successful articulatory execution. The fact that AWS can sometimes produce certain motor programs while they cannot at other times (Tichenor and Yaruss, 2021), makes it difficult to understand which motor programs or combinations thereof, are more likely to produce stuttering events. One attempt that has gained considerable momentum in this regard is the work by Howell and colleagues (Al-Tamimi et al., 2013; Dworzynski and Howell, 2004; Howell and Au-Yeung, 2007; Howell et al., 2006) which shows that higher phonetic complexity (for IPC scores of the current materials see Methods section) can be associated with higher stuttering rates in adults. However, no clear pattern has yet emerged that can reliably predict stuttering events and provide insights into underlying compensatory strategies that could be successfully applied in treatment approaches. In

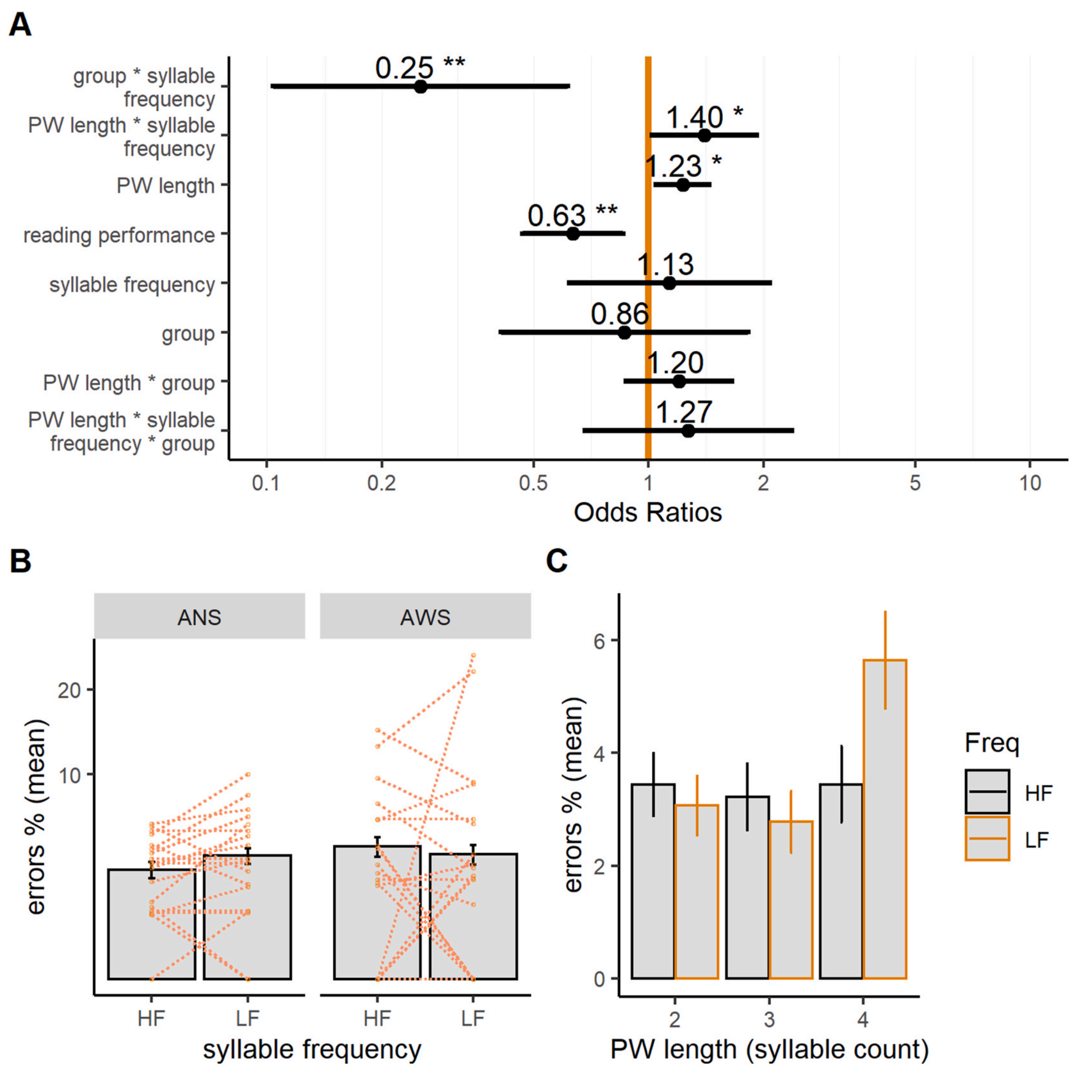

Fig. 4. Probability of errors in initial syllable position ($n_{AWS} = 19$, $n_{FC} = 19$). Note. A. Probability of errors in initial syllable position as indicated by Odds ratios was significantly influenced by the interaction between syllable frequency and group, by syllable frequency and pseudoword (PW) length, by pseudoword length and by reading skill. B. Bar plots represent group means of errors for high (HF)- and low (LF)-frequency syllables. C. Bar plots represent means of errors from both groups for two-, three-, and four-syllabic pseudowords separated by syllable frequency. Error bars represent standard errors.

the current study, error analyses did not reveal any general or individual patterns in our AWS group. Our stimuli were carefully controlled for many potentially confounding factors (see Materials). However, we did not control for phonemic bigram frequency or vowel length. These factors have previously been found to influence syllable frequency effects (e.g., Coalson et al., 2019; Kalveram, 2001). When we included these factors as predictors in our analyses, none of them reached significance. Nevertheless, future studies should control for or directly test these factors to understand the underlying error patterns of AWS and potential compensatory strategies.

4.2. Effects of pseudoword length

In the current study, there was a clear length effect on stuttering rates in AWS and on error rates over all syllables in both speaker groups: disyllabic pseudowords were produced more fluently and more accurately than three- and four-syllabic pseudowords. This finding is consistent with results showing that shorter words are produced faster and more accurately than longer words in fluent speakers (e.g., Damian & Dumay, 2007; Meyer et al., 2003; Windsor et al., 2010) in AWS (Byrd et al., 2012, 2015; Sasisekaran and Weisberg, 2014), and in speakers with neurological language disorders (Croot et al., 1998; Nickels and Howard, 2004). In support of our findings, previous studies have reported that adults and children who stutter are more likely to stutter on long words than on short words (e.g., Logan and Conture, 1995; Max et al., 2019) and that children who stutter produce more disfluencies on long pseudowords than on short pseudowords (Sasisekaran and Weathers, 2019).

In the psycholinguistic literature, effects of word length have been discussed in terms of what constitutes an appropriate articulatory unit and how the execution of that unit is temporally coordinated with respect to the larger planning unit of which that unit may be a part. In other words, following the incremental nature of speech planning, the first syllable of a multisyllabic word may already be phonetically encoded and be fed into the articulatory network before subsequent syllables are ready for articulation. The decision of whether to store the first syllable in the articulatory buffer to wait for its successors or execute it immediately depends i) on the assumption that speakers have a *fixed* planning unit for articulation (e.g., a single segment, see for example, Dell, 1993 or an entire (phonological) word, see for example, Levelt and Wheeldon, 1994), whereby articulation is initiated independently of the status of later parts of an utterance. (ii) Alternatively, speakers have a *flexible* planning unit for articulation and can coordinate, i.e., decide within a given context on what basis to initiate articulation. This decision is likely to depend on the encoding status of immediately following elements to ensure a steady flow of speech, taking into account factors such as the integrity of linguistic structures and listener-oriented speech. Moreover, previous research that has systematically tested the interplay between articulation initiation, planning scope (Alario et al., 2002a, 2002b; Levelt, 2002) and other factors has found that the temporal coordination of planning and execution is driven by utterance format (e.g., Meyer et al., 2007), but is also language (i.e., by language-specific stress patterns, syllabic transparency, Cholin et al., 2011) and speaker dependent (i.e., 'hasty' vs. 'careful' speakers, Schriefers, 1999). Thus, while a number of contributing factors have been identified, their interplay is still largely unknown. Importantly, it has been argued that for the first syllable to be processed beyond phonetic encoding, the entire phonological word to which it belongs must be fully syllabified so that stress and other timing parameters can be assigned (e.g., Damian et al., 2010; Levelt et al., 1999).

Clear length effects will only be found if speakers reliably collect all the syllables belonging to a phonological word before articulation is initiated. This is what has been observed in studies where participants seem to have adopted specific response criteria to respond to separate blocks of short or long items (e.g., Meyer et al., 2003).

In the current task, participants were presented with length-heterogeneous blocks and they were instructed to start articulation only when they felt the urge to speak. Thus, without the pressure to produce the pseudowords as fast as possible, participants may have chosen to encode all syllables before speech onset. In fact, this encoding strategy may explain why we observed length effects across both speaker groups in our main analyses (AWS and FC).

Interestingly, our results show that word length during speech planning processes seems to affect the accuracy of initial syllables: in both groups, error rates on first syllables increased with pseudoword length. A similar pattern of errors was observed in a recent study investigating memory storage and inter-articulator coordination for 1-syllabic (experiment 1) and 2-syllabic novel sequences (Masapollo et al., 2023). Comparing the error rates of these two experiments, this research group found that the speech accuracy of the initial syllable in Experiment 2 decreased compared to Experiment 1, possibly as a result of having planned and stored the additional second syllable.

A closer inspection of our results further shows that with increasing length the prevalence of an error in initial syllable position decreases whereas it increases for the second or third/fourth position (see Supplementary Fig. 1B and C). Masapollo et al. (2023) suggested that more errors on the second syllable in two-syllabic pseudowords may reflect a primacy effect, i.e., the concurrent storage and production of the first syllable leads to a decreased performance of the second syllable. Although we did not investigate the role of prosody on errors, the prosodic structure of the spoken pseudowords may also have influenced the prevalence of errors in different syllable positions. First evidence shows that across sentences speech errors occur more often in both the early and late-mid positions than in the initial positions of an intonational boundary (Choe and Redford, 2012). Furthermore, Croot et al. (2010) showed that the way in which a sequence of 4 one-syllabic words was emphasized influenced the occurrence of error rates within the sequence.

In AWS, there was a parallel effect on stuttering rates in first position: increasing pseudoword length led to an increase in stuttering of initial syllables. However, for four-syllabic pseudowords, in contrast to the findings for speech errors, the majority of stuttering events in AWS remained on the initial syllables (Supplementary Fig. 1A). Thus, although the higher load on the speech planning system increased stuttering rates, we interpret the remainder of stuttering events in initial syllables, regardless of word length, as evidence that stuttering is caused by a different (patho)mechanism than speech errors.

4.3. *Interaction of pseudoword length and syllable frequency*

How can we interpret the significant two-way interaction between pseudoword length and syllable frequency for errors in initial syllables? We suggest that this significant interaction together with the descriptive results could imply that the increased speech planning load due to four-syllabic pseudowords with a low-frequency initial syllable may mask the effects of the inversed syllable frequency effect in AWS. In AWS, the segment-by-segment construction of the low-frequency initial syllable, which consisted of 2–4 phonemes, may have added additional load to the planning of a 4-syllabic pseudoword. Previous studies on length effects in AWS have reported that group differences in error rates only begin to emerge in much longer words, i.e., words with 6–7 syllables (Byrd et al., 2012, 2015; Sasisekaran and Weisberg, 2014). The results suggest that the speech accuracy of AWS seems to be affected more by length effects than by syllable frequency.

4.4. *Neural correlates of syllable frequency effects in stuttering*

The findings of the current study support the dual-route account, suggesting qualitatively different neural processing. Previous evidence from neuroimaging studies in FC, let alone in AWS, is far from clear. Masapollo et al. (2021) investigated whether AWS differed from FC in their BOLD activation during speech motor learning of pseudowords containing non-native consonant clusters. Irrespective of the group, Masapollo et al. (2021) replicated the findings of Segawa et al. (2015): during the production of novel sequences left hemispheric activity was higher in regions of the speech motor network when compared to the production of practiced sequences. This resembles differences in brain activity for high and low frequency syllables (Carreiras et al., 2006; Papoutsi et al., 2009; Tremblay et al., 2016). However, these results cannot speak to our specific assumptions about the interplay between the two routes in AWS. One could speculate that frequency specific neural differences in AWS and FC evolve in very narrow time windows that cannot be measured by an infra-slow BOLD activity. Indeed, EEG studies on FC investigating syllable frequency effects have observed different ERP amplitudes within time windows lasting less than 100ms before the onset of articulation (Bürki et al., 2015, 2020). An analysis of this syllable frequency effect in AWS' EEG recordings is currently underway.

Noteworthy, a computational simulation of the GODIVA (Gradient Order Directions in Velocities of Articulator) model showed that stuttering could result from increased dopaminergic levels within the basal ganglia and/or a white matter impairment leading to a delayed motor program selection (Civier et al., 2013). Unfortunately, the development of motor programs or syllable frequency effects have not yet been simulated within an adapted model of the speech production system specific to DS. With the DIVA respectively GODIVA model, which provide a neurocomputational implementation of the development of motor programs resp. the syllable frequency effect, such a simulation may even be possible in stuttering (Bohland et al., 2010; Meier and Guenther, 2023) and could offer interesting insights.

The association between the basal ganglia including the putamen and their role in sequential motor learning and the emergence of stuttering has been addressed by Alm (2004). More recent neurological findings of abnormal functional connectivity (Lu et al., 2010b) and reduced volume (Beal et al., 2013) of the putamen in children who stutter together with findings of increased activity of the putamen during motor learning tasks (Hardwick et al., 2013; Orban et al., 2010) could support the notion of impaired automaticity of syllables. The theory of less automatized speech programs has been also addressed by (Alm, 2021) linking evidence of increased fluency in persons who stutter when speaking without using the automatic mode of speech production and the role of basal ganglia for processing automatized movements. A recent survey on former stuttering persons showed, that individuals attribute their recovery from stuttering to methods requiring speakers to consciously manipulate their speech production (Neumann et al., 2019). In addition, speaking with an external focus rather than an internal focus increased flexibility in AWS (Bauerly and Jackson, 2024).

4.5. Limitations

The current study measured speech fluency by analysing stuttering events and errors. This measure is susceptible to perceptual bias, i.e., more salient speech errors are more easily recognized by human perception and could therefore be over-represented in the analysis (Alderete and Davies, 2019). Applying this concept to stuttering symptoms, it could be argued that less salient stuttering events, for example, those that are very short or produced more fluently, are harder to identify. We tried to minimize perceptual bias by using offline analyses (Alderete and Davies, 2019) such that the raters relistened the pseudowords prior to transcription. Further, the raters were trained and used a standardized protocol to ensure consistency. However, only an objective observation of speech such as kinematic measures could avoid perceptual bias. Several kinematic studies on stuttering have reported a higher variability of articulatory movements in persons who stutter compared to FC (e.g. Smith et al., 2010. To our knowledge kinematic measures of AWS investigating syllable frequency effects do not exist. Moreover, it is important to note that, while the psycholinguistic model of word production by Levelt and colleagues (Levelt et al., 1999) and the neurocomputational models by Guenther and colleagues (Guenther, 2016) capture different stages of the production process (conceptualisation to phonological/phonetic encoding and phonological/phonetic encoding to articulation, respectively), neither model provides a cohesive overview of the entire process. Accounts of speech motor execution with particular focus on the low-level control of the speech articulators (Parrell et al., 2019), may also explain the observed effects.

5. Conclusion

This study revealed how syllable frequency as a proxy for automaticity and pseudoword length as a proxy for motoric complexity jointly shape speech planning and control. For short words, the motor planning load is low, so processes such as competitive activation among strongly represented syllables may dominate. AWS may have difficulties inhibiting or selecting among these high-frequency motor patterns, resulting in substitution or selection errors and reflecting inefficient access to stored motor representations. In contrast, long pseudowords composed of initial low-frequency syllables elicited more errors. Because these syllables are weakly represented, increasing word length amplifies planning and sequencing demands. Consequently, AWS appear to struggle to retrieve and maintain less automatic representations over extended planning intervals, suggesting instability or decay of phonological–motor memory traces during prolonged speech planning. This aligns with models proposing that stuttering involves an impaired integration between phonological retrieval and motor planning, and that AWS have reduced temporal stability of phonological–motor representations especially as utterances get longer. Future studies are needed to investigate the neurophysiological basis and the interaction with other linguistic factors.

CRediT authorship contribution statement

Alexandra Korzeczek: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Nicole E. Neef:** Writing – review & editing, Formal analysis. **Jana Wiechmann:** Writing – review & editing, Formal analysis. **Katharina Hammers:** Writing – review & editing, Formal analysis. **Arno Olthoff:** Writing – review & editing, Resources, Methodology. **Martin Sommer:** Writing – review & editing, Supervision, Resources, Conceptualization. **Joana Cholin:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.neuropsychologia.2025.109295.

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